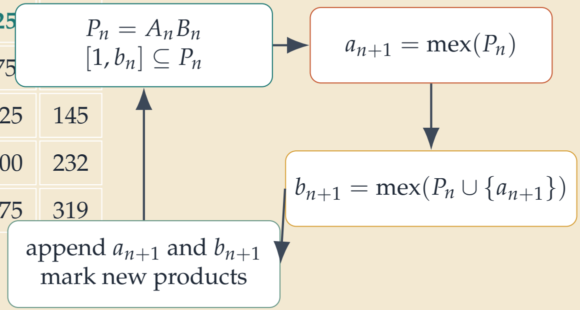
Prime-separator frontiers and the interleaving theorem

A. Published 5×11 prefix

1	**2**	**4**	**7**	**9**	**13**	**15**	**18**	**23**	**25**	[illegible]
3	6	12	21	27	39	45	54	69	75	[illegible]
5	10	20	35	45	65	75	90	115	125	145
8	16	32	56	72	104	120	144	184	200	232
11	22	44	77	99	143	165	198	253	275	319

Border terms are shown in teal. Every interior entry is a product $b_i a_j$, so the finite square is completely determined by the first row and first column.

B. One update step



C. Proven order of the borders

The key inductive invariant is the frontier coverage statement $[1, b_n] \subseteq P_n$. It forces strict interleaving: $a_n < b_n < a_{n+1} < b_{n+1}$ for every $n \geq 2$. The empirically observed axis-1 marker therefore has a formal explanation: the unique border term inside the row gap (a_n, a_{n+1}) is always b_n itself.